HTML HW4

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By definition of the Hessian we have

$$A_E(\mathbf{w})_{ij} = \frac{\partial^2}{\partial w_i \partial w_j} E_{in}(\mathbf{w})$$

$$\frac{\partial E_{in}(\mathbf{w})}{\partial w_i} = \frac{1}{N}\sum_{n=1}^{N} \frac{\exp(-y_n \mathbf{w}^T \mathbf{x}_n)(-y_n x_{n_i})}{1+\exp(-y_n w^T \mathbf{x}_n)}$$

$$\begin{aligned}
\frac{\partial}{\partial w_j}\left(\frac{\partial E_{in}(\mathbf{w})}{\partial w_i}\right) &= \frac{1}{N}\sum_{n=1}^{N} \frac{(\exp(-y_n \mathbf{w}^T \mathbf{x}_n)(\mathbf{x}_{n_i}\mathbf{x}_{n_j}))(1+\exp(-y_n \mathbf{w}^T \mathbf{x}_n)) - \exp(-y_n \mathbf{w}^T \mathbf{x}_n)^2 (x_{n_i} x_{n_j})}{(1+\exp(-y_n \mathbf{w}^T \mathbf{x}_n))^2} \\
&= \frac{1}{N}\sum_{n=1}^{N} \frac{\exp(-y_n \mathbf{w}^T \mathbf{x}_n)}{(1+\exp(-y_n \mathbf{w}^T \mathbf{x}_n))^2} x_{n_i} x_{n_j} \\
&= \frac{1}{N}\sum_{n=1}^{N} \frac{1}{1+\exp(-y_n \mathbf{w}^T \mathbf{x}_n)} \frac{1}{1+\exp(y_n \mathbf{x}^T \mathbf{x}_n)} x_{n_i} x_{n_j} \\
&= \frac{1}{N}\sum_{n=1}^{N} h_t(\mathbf{x}_n) h_t(-\mathbf{x}_n) x_{n_i} x_{n_j}
\end{aligned}$$

(x_{n_i} is the i-th element of the $\mathbf{x}_n$, and is the same as X_{ji}) For a N by d matrix X and a N by N diagonal matrix D, we have

$$(X^T D X)_{ij} = \sum_{n=1}^{N} D_{nn} X_{ni} X_{nj}$$

By comparing the expressions, we can obtain

$$D_{ij} = \begin{cases} 0 & i \neq j \\ \frac{1}{N} h_t(\mathbf{x}_n) h_t(-\mathbf{x}_n) & i = j \end{cases}$$

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In this problem, we use w_{j_i} to denote the i-th element of $\mathbf{w}_j$, which is the same as W_{ij}
Since E_{in} is minmized with SGD, we know that

$$\mathrm{V}_{ij} = -\frac{\partial}{\partial w_{j_i}} \text{err}(W, \mathbf{x}, y)$$

$$\begin{aligned}
\frac{\partial}{\partial w_{j_i}} \text{err}(W, \mathbf{x}, y) &= -\frac{1}{h_y(\mathbf{x})} \frac{\partial h_y(\mathbf{x})}{\partial w_{j_i}} \\
&= -\frac{1}{h_y(\mathbf{x})} \frac{[\![y = i]\!] \exp\big(\mathbf{w}_y^T \mathbf{x}\big)(\sum_{k=1}^K \exp(\mathbf{w}_k^T \mathbf{x}))x_j - \exp\big(\mathbf{w}_y^T \mathbf{x}\big) \exp\big(\mathbf{w}_i^T \mathbf{x}\big) x_j}{\Big(\sum_{k=1}^K \exp\big(\mathbf{w}_k^T \mathbf{x}\big)\Big)^2} \\
&= -\frac{1}{h_y(\mathbf{x})} \frac{([\![y = i]\!] - \exp\big(\mathbf{w}_i^T \mathbf{x}\big)) \exp\big(\mathbf{w}_y^T \mathbf{x}\big) x_j}{(\sum_{k=1}^K \exp\big(\mathbf{w}_k^T \mathbf{x}\big))^2} \\
&= -([\![y = i]\!] - \frac{\exp\big(\mathbf{w}_i^T \mathbf{x}\big)}{\sum_{k=1}^K \exp\big(\mathbf{w}_k^T \mathbf{x}\big)}) x_j \\
&= (h_i(\mathbf{x}) - [\![y = i]\!]) x_j
\end{aligned}$$

Notice that

$$(\mathbf{x}_n \cdot \mathbf{u}^T)_{ij} = x_j u_i$$

Hence by comparing the expressions, $\mathbf{u}$ is given by

$$u_i = -h_i(\mathbf{x}_n) + [\![y_n = i]\!]$$

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For MLR , we know that $\sum_{n=1}^{N} \mathrm{err}(\mathrm{W}, \mathbf{x}, y)$ is minimized at its optimal solution, $(\mathbf{w}_1^\star, \mathbf{w}_2^\star)$. Also

$$\begin{aligned}
\sum_{n=1}^{N} \mathrm{err}(\mathrm{W}, \mathbf{w}, y) &= \sum_{n=1}^{N} \ln\left(\frac{\exp\left(\mathbf{w}_{y_n}^T \mathbf{x}_n\right)}{\exp(\mathbf{w}_1^T \mathbf{x}_n) + \exp(\mathbf{w}_2^T \mathbf{x}_n)}\right) \\
&= \begin{cases} -\sum_{n=1}^{N} \ln\left(1 + \exp((\mathbf{w}_2^T - \mathbf{w}_1^T)\mathbf{x}_n)\right) & \text{if } y_n = 1, \\ -\sum_{n=1}^{N} \ln\left(1 + \exp((\mathbf{w}_1^T - \mathbf{w}_2^T)\mathbf{x}_n)\right) & \text{if } y_n = 2 \end{cases} \\
&= -\sum_{n=1}^{N} \ln\left(1 + \exp(-y_n'(\mathbf{w}_2 - \mathbf{w}_1)^T \mathbf{x}_n)\right)
\end{aligned}$$

For logistic regression

$$E_{\mathrm{in}}(\mathbf{w}) = -\frac{1}{N} \sum_{n=1}^{N} \ln\left(1 + \exp(-y_n' \mathbf{w}^T \mathbf{x}_n)\right)$$

By comparing the expressions, we see that $E_{\mathrm{in}}(\mathbf{w})$ is minimized with

$$\mathbf{w}_{lr} = \mathbf{w}_2^\star - \mathbf{w}_1^\star$$

Hence this is the optimal solution to the logistic regression.

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The linear hypothesis that mimizes E_{in} is the line that passes $(x_1, f(x_1))$ and $(x_2, f(x_2))$. Hence

$$\begin{aligned} g(x) &= f(x_1) + \frac{f(x_2) - f(x_1)}{x_2 - x_1}(x - x_1) \\ &= -2(x_1 + x_2)x + 2x_1x_2 + 1 \end{aligned}$$

For such a hypothesis, $E_{\text{in}} = 0$.
E_{out} is given by

$$\begin{aligned} E_{\text{out}} &= \int_0^1 (g(x) - f(x))^2 \, \mathrm{d}x \\ &= \int_0^1 \left(2x_1x_2 - 2(x_1 + x_2)x + 2x^2\right)^2 \mathrm{d}x \end{aligned}$$

$$\begin{aligned} \mathbb{E}_{\mathcal{D}}(|E_{\text{in}} - E_{\text{out}}|) &= \int_0^1 \int_0^1 |E_{\text{out}}| \, \mathrm{d}x_1 \, \mathrm{d}x_2 \quad (E_{\text{in}} = 0) \\ &= \int_0^1 \int_0^1 \int_0^1 \left(2x_1x_2 - 2(x_1 + x_2)x + 2x^2\right)^2 \mathrm{d}x \, \mathrm{d}x_1 \, \mathrm{d}x_2 \\ &= \frac{2}{15} \end{aligned}$$

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Let

$$\tilde{\mathrm{X}} = \mathrm{X} + \mathrm{E}$$

Where

$$\mathrm{E} = \begin{bmatrix} | & \dots & | \\ \epsilon & \dots & \epsilon \\ | & \dots & | \end{bmatrix}$$

$$\begin{aligned} \mathrm{X}_h^T \mathrm{X}_h &= \begin{bmatrix} \mathrm{X}^T & \tilde{\mathrm{X}}^T \end{bmatrix} \begin{bmatrix} \mathrm{X} \\ \tilde{X} \end{bmatrix} \\ &= \mathrm{X}^T \mathrm{X} + \tilde{\mathrm{X}}^T \tilde{\mathrm{X}} \\ &= 2\mathrm{X}^T \mathrm{X} + \mathrm{E}^T \mathrm{X} + \mathrm{X}^T \mathrm{E} + \mathrm{E}^T \mathrm{E} \end{aligned}$$

Therefore

$$\begin{aligned} \mathbb{E}[\mathrm{X}_h^T \mathrm{X}_h] &= \mathbb{E}[2\mathrm{X}^T \mathrm{X} + \mathrm{E}^T \mathrm{X} + \mathrm{X}^T \mathrm{E} + \mathrm{E}^T \mathrm{E}] \\ &= 2\mathrm{X}^T \mathrm{X} + \mathbb{E}[\mathrm{E}^T]\mathrm{X} + \mathrm{X}^T \mathbb{E}[\mathrm{E}] + \mathbb{E}[\mathrm{E}^T \mathrm{E}] \end{aligned}$$

Since ϵ is generated i.i.d. from a normal distribution with variance σ^2, we have

$$\begin{aligned} \mathbb{E}[\mathrm{E}^T] &= \mathrm{O}_{(d+1)\times N} \\ \mathbb{E}[\mathrm{E}] &= \mathrm{O}_{N\times(d+1)} \\ \mathbb{E}[\mathrm{E}^T \mathrm{E}] &= N\sigma^2 \mathrm{I}_{d+1} \end{aligned}$$

Hence

$$\mathbb{E}[\mathrm{X}_h^T \mathrm{X}_h] = 2\mathrm{X}^T \mathrm{X} + N\sigma^2 \mathrm{I}_{d+1}$$

So $(\alpha, \beta) = (2, N)$